

ETP formalization benchmark · two-stage at scale, no-think, + GPT-5.5

Stories generated from Equational Theories Project implications (`storyform.py`), each abstracted by the model into a literal description (`abstract_prompt.md`) and then formalized in a second call (`literal_prompt.md`) over OpenRouter, and graded syntactically (`checkform.py`). Correct = exact, or an accepted symmetry (sides swapped, or both equations uniformly dualized).

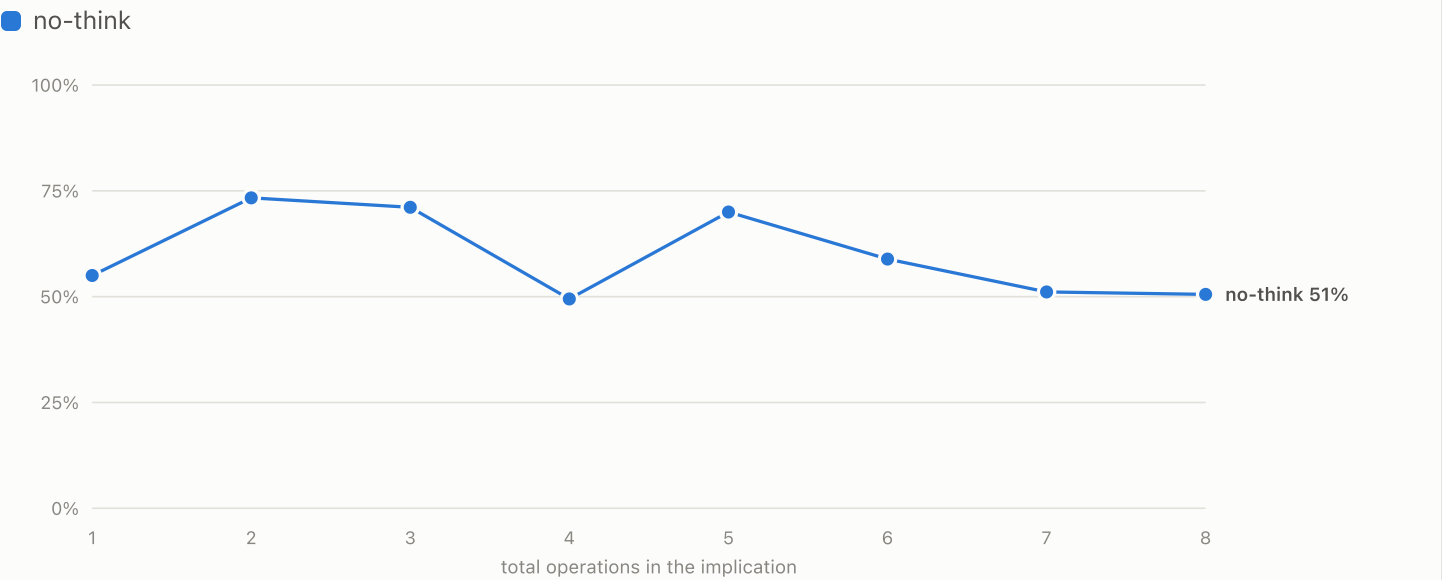
experiments/07-two-stage-scale/runs/run-strat20-s0-think-off

Runs	Models	Stories	Graded calls	Overall correct
1	9	160	1,440	60%

160 pairs, 20 per operation-count bin 1–8 · seed 0 · no-think

FIGURE 1 Accuracy vs operation count — stratified 160

Correct rate pooled over all models, by the implication's total operation count (both equations combined). Sample: 160 pairs, 20 per total-operation bin 1–8 — the full complexity range, including trivial laws.

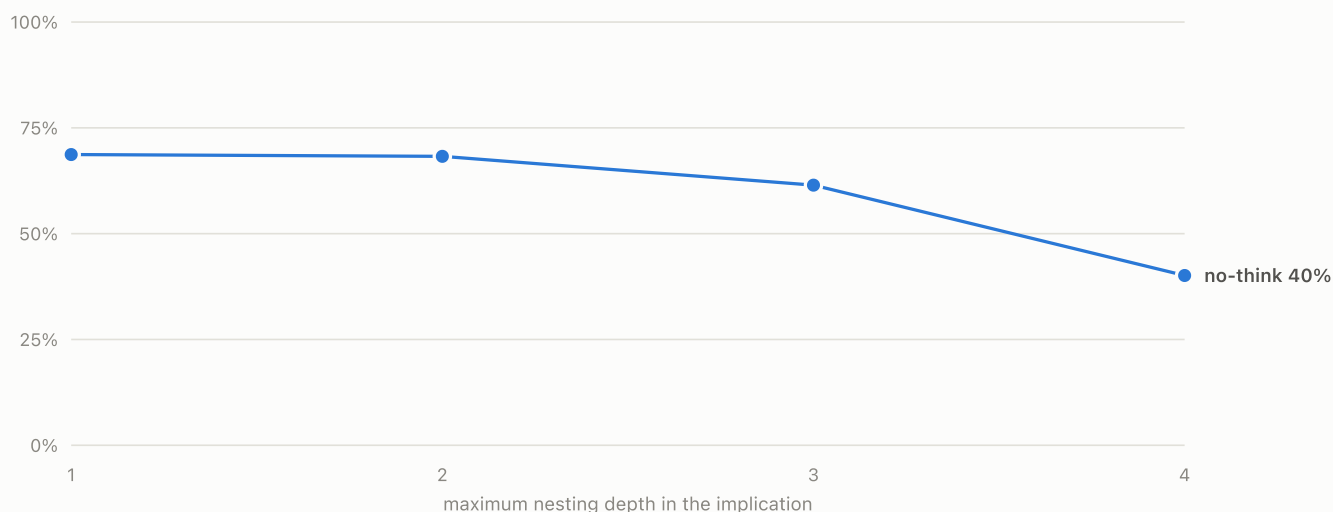


► Data: pooled correct rate per operation-count bin

FIGURE 2 **Accuracy vs nesting depth — stratified 160**

Correct rate pooled over all models, by the deepest operation nesting across the four equation sides. Sample: 160 pairs, 20 per total-operation bin 1–8 — the full complexity range, including trivial laws.

■ no-think

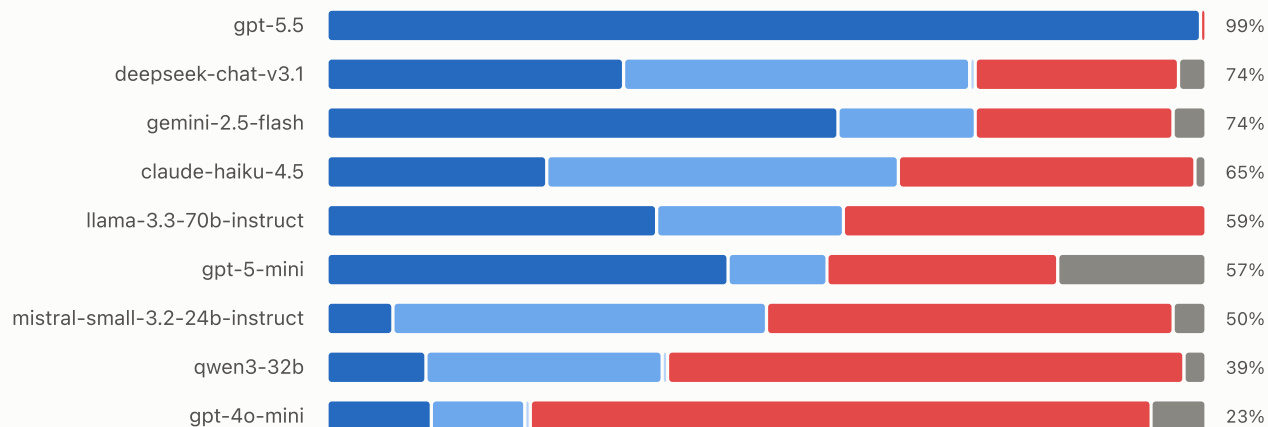


► Data: pooled correct rate per depth bin

FIGURE 3 **Verdict composition — no-think regime · stratified 160**

How each model's answers grade out; the right-hand figure is the correct rate (exact + swapped + dualized). Sample: 160 pairs, 20 per total-operation bin 1–8 — the full complexity range, including trivial laws.

■ exact ■ correct-swapped ■ correct-dualized ■ wrong ■ unparseable



► Data: verdict counts per model